

OASIS INFOBYTE – Data Analytics Task 1

Project Title

Retail Sales Exploratory Data Analysis

Objective

The objective of this project is to analyze retail sales data and identify useful patterns and insights related to products, customers, sales, and purchasing behavior.

Tools & Technologies

- Python
- Pandas
- Matplotlib
- Jupyter Notebook

Dataset

The dataset contains retail transaction information including:

- Transaction ID
- Date
- Customer ID
- Gender
- Age
- Product Category
- Product
- Quantity
- Unit Price
- Total Sales
- Payment Method
- City

Data Cleaning

The following preprocessing steps were performed: 1. Checked the dataset structure and information. 2. Identified duplicate records. 3. Removed duplicate records. 4. Checked for missing values. 5. Filled missing Age values using the median. 6. Filled missing Payment Method values using the most frequent value.

Exploratory Data Analysis

The analysis includes:

- Total and average sales
- Total quantity sold
- Average customer age
- Sales by product category
- Sales trends over time
- Sales by gender
- Best-performing product categories
- Top-selling products

Visualizations

Three visualizations were created: 1. Total Sales by Product Category 2. Sales Over Time 3. Total Sales by Gender

Conclusion

The retail sales data was successfully cleaned and analyzed using Python. The analysis and visualizations provide insights into sales performance, customer purchasing patterns, product categories, and sales trends.